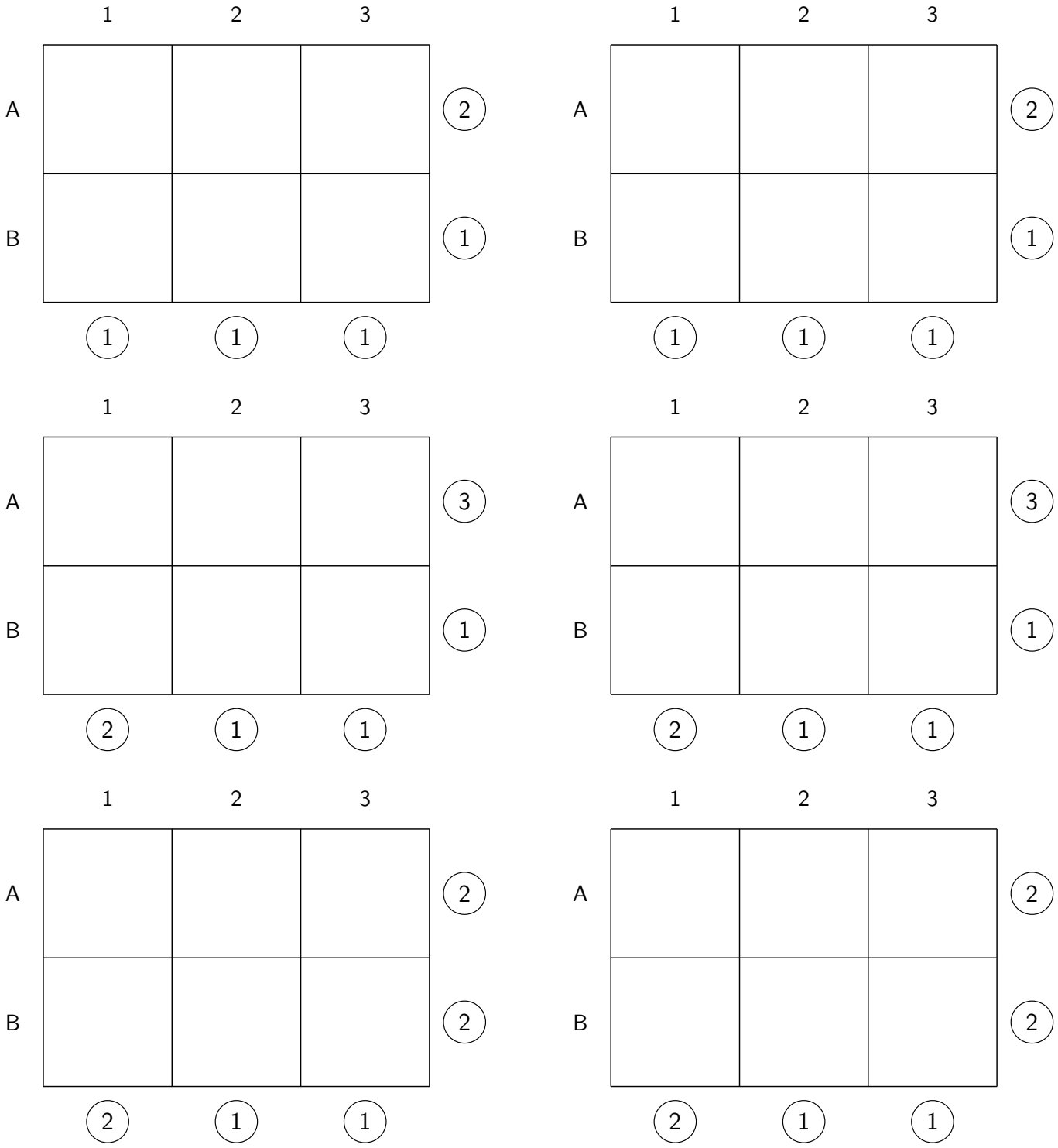
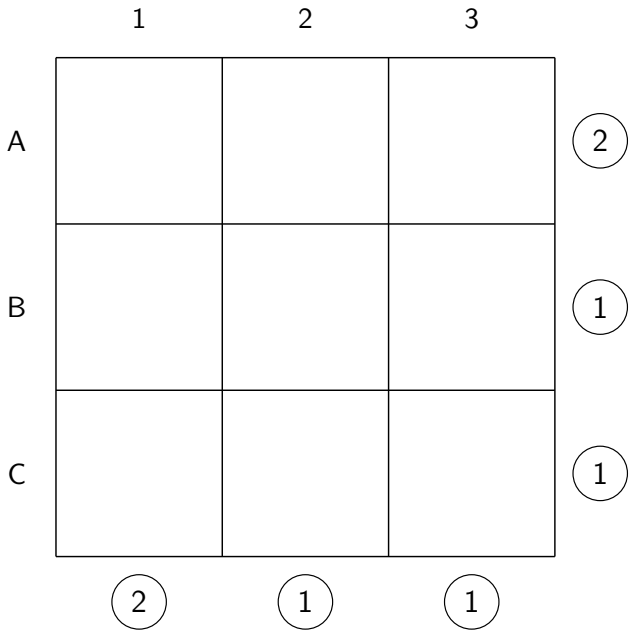
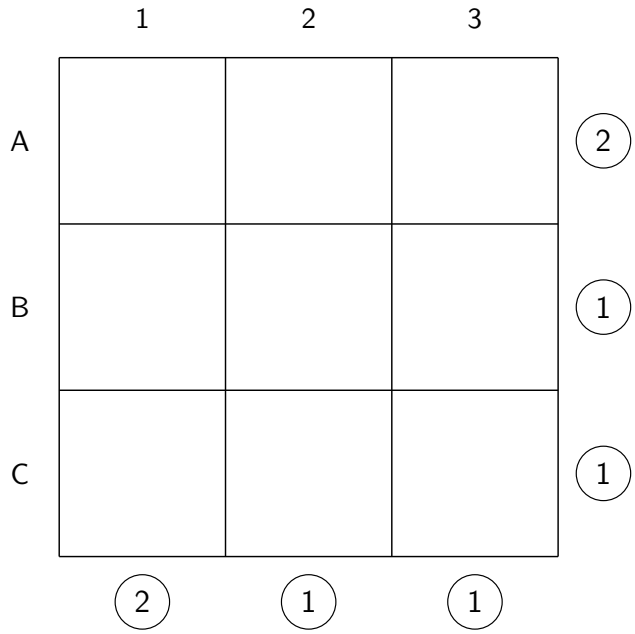
Each circled number counts the counters in its whole row or column. Put at most one counter in a cell. Two pictures are different if any cell is different.

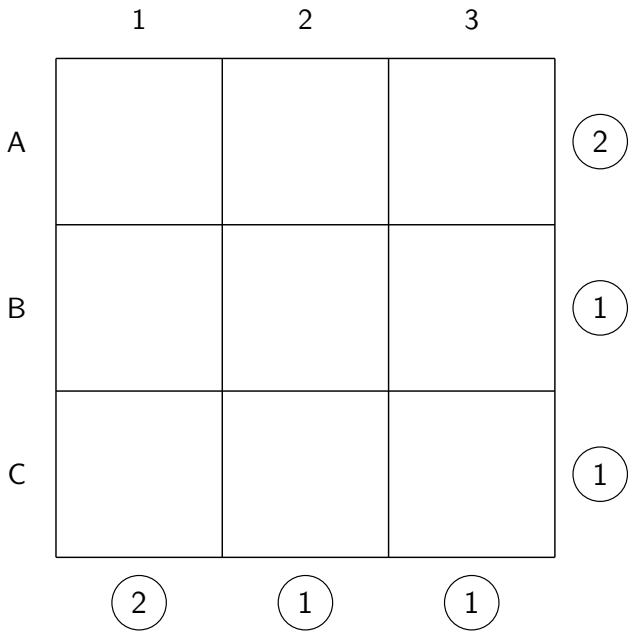
**Problem 1:** For each pair of grids, make two different pictures that match the counts, if you can.

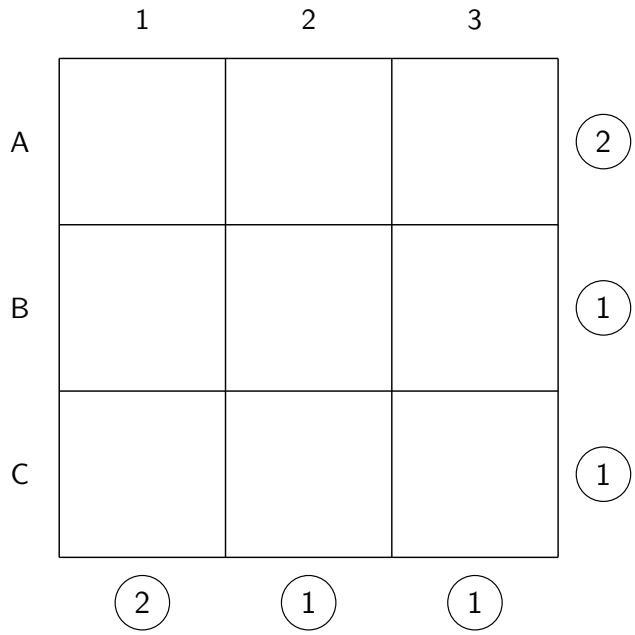


**Problem 2:** Find and draw every different picture with these counts. How can you be sure your list is complete?

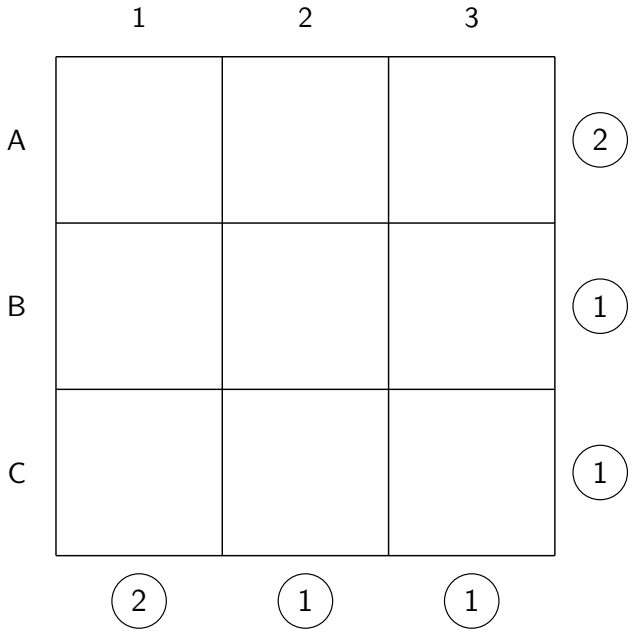


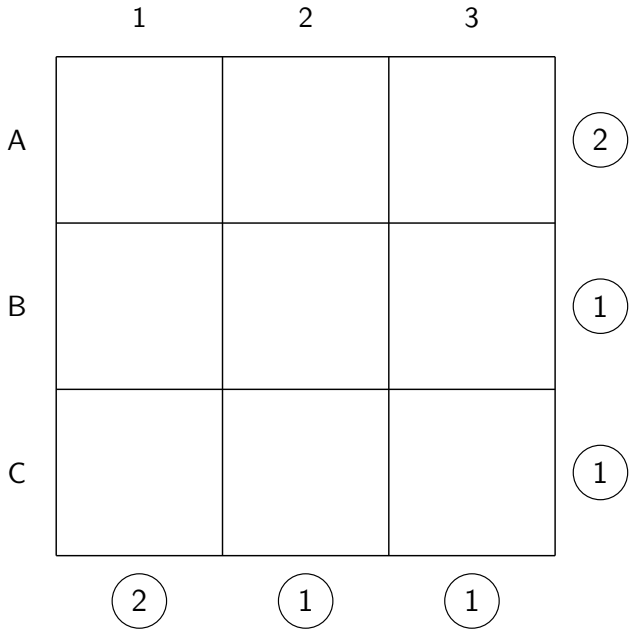



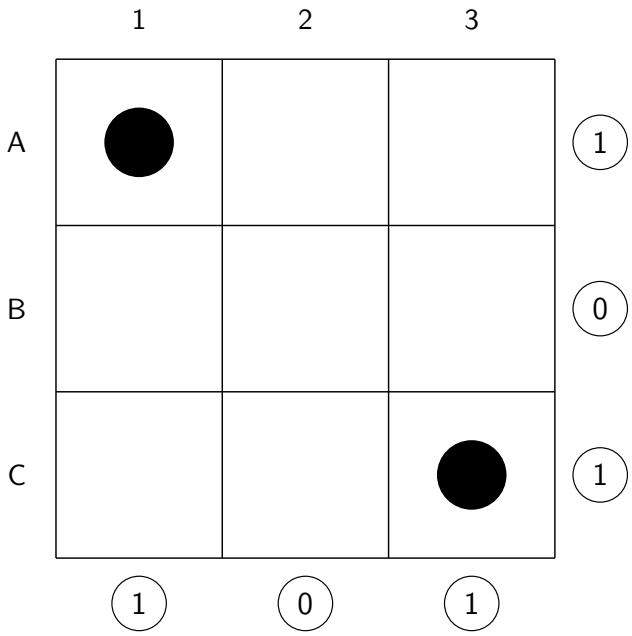



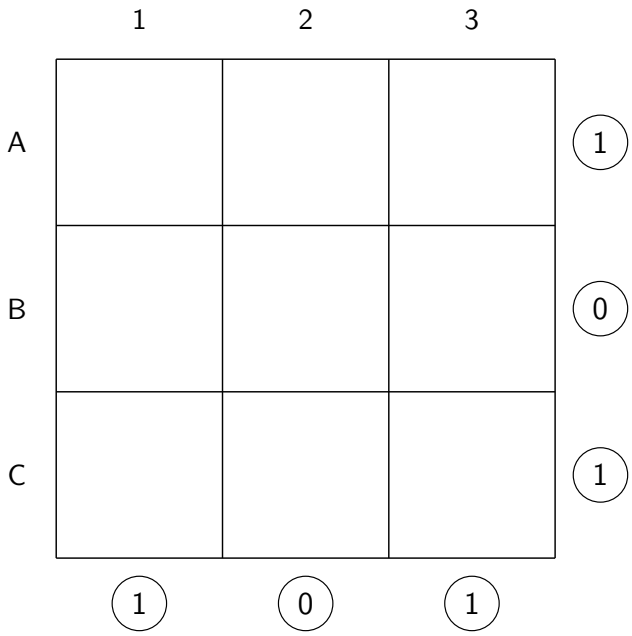
## Problem 2 (continued)




**Problem 3:** Move exactly two counters from the filled picture to make a different picture with the same counts.



**Problem 4:** A switch moves the two counters to the empty corners shown; every other cell stays fixed. The corners may use any two rows and columns. Find every switch in each picture and draw its rectangle. Which pictures have none?

One switch:

